

Kayané ROBACH

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<https://kayanerobach.github.io/>

WORK EXPERIENCE

2022 - 2026 Statistical Consultant, Medical Research; Amsterdam UMC, The Netherlands

Advised clinical researchers across internal medicine, ICU, infectious diseases on study design and analysis (mixed models, regression, weighting schemes, causal discovery) turning ambiguous research questions into executable statistical plans, reviewing and correcting analyses pre-publication across small studies (50-500 observations).

2022 - 2026 Teaching Assistant, Statistics & R Programming; Vrije Universiteit Amsterdam, The Netherlands

Taught statistical and epidemiological reasoning to Bachelor students and coached PhD candidates on modelling methods in R.

2022 Research Assistant, Random Matrix Theory & Theoretical Ecology; LIGM, Paris, France

Formalised feasibility and stability problems for Lotka-Volterra systems in large dimension into a phase transition detection algorithm.

2021 Data Scientist; 55, Paris, France

Analysed A/B tests and survey data to identify cognitive drivers of online user consent decisions, converting UX and compliance problematics into behavioural models with design recommendations. Contributed agent-based simulations with reinforcement learning to model population behaviour and test marketing strategies pre-deployment.

2020 Data Scientist; EvidenceB, Paris, France

Built pipelines converting raw clickstream from an AI powered learning platform into clean data. Applied clustering, dimension reduction techniques, recommendation algorithms, to diagnose drop-out and personalise content sequencing. Turned questions into analysis plans and actionable recommendations for the product and its data collection pipeline.

EDUCATION

2022 - 2026 PhD in Mathematical Statistics; Amsterdam UMC, The Netherlands

Causal Record Linkage *Developed and implemented novel causal inference and data fusion methods to merge heterogeneous data sources. Published 2 peer reviewed articles (JRSSSC, Statistics in Medicine); 2 further articles under review. Presented at international conferences (ICSIDS, Eurocim, CompStats, IBS, IBC). Automated software maintenance workflows using agentic coding tools. Organised the Department annual outing for 100 people (planning, budget, logistics, communication).*

2021 - 2022 Master 2 Mathematics and Applications; ENS Paris-Saclay, France

MVA *Architecture design, implementation, training of models for computer vision (CNN, bag-of-features, gradient based learning); advanced learning methods for text and graph data (NLP, graph-based text mining, knowledge distillation); generative models, deep reinforcement learning models, RNN, LSTM, GAN, transformers; topological data analysis, computational statistics, geometry of shape space, probabilistic graphical models, kernel methods*

2020 - 2021 Master 1 Applied Mathematics and Statistics; Toulouse School of Economics, France

Mathematics and Economic Decision *Games and equilibria, Probability modelling, Strategic and dynamic optimisation, Time series, Martingales theory, Advanced functional analysis*

2019 - 2022 Magister Econometrics and Statistics; Toulouse School of Economics, France

Economist Statistician *Machine Learning, Stochastic simulation, Markov chains and applications, Data analysis, Programming, Statistics, Optimisation for big data*

2017 - 2020 Double Degree Economics and Mathematics; Toulouse School of Economics, France

Economics, Mathematics and Computer Science applied to human and social sciences *Statistics, Measure theory and probability, Algebra, Algorithmic, Micro and Macro Economics, Analysis and optimisation, Econometrics*

OUTREACH

2025 - now CoMeEcon Seminar; VVS-OR, The Netherlands *Manage funds, program operations, communications and logistics of the biannual seminar on computational statistics.*

2024 - now Fietsenwerkplaats Smerig; Amsterdam, The Netherlands *Volunteer work; build and repair bicycles; assemble and disassemble bikes, wheels, internal hub mechanisms.*

MISCELLANEOUS

Tools: R, C++, SAS, STATA, SQL, Python (PyTorch, Keras, TensorFlow), llama.cpp, opencode
French (native), English (fluent), Dutch (intermediate), Spanish (intermediate)
Piano & Violin amateur player, amateur sailor